
KANISHKA KOPPERLA

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SKILLS

Programming & Languages: Python, SQL, NoSQL

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, Hugging Face, LLMs, NLP, Transformers, RAG, OpenCV

Cloud & Data Platforms: Docker, Apache Kafka, Google Pub/Sub, GCP (BigQuery, Cloud Functions), GitHub Actions

Backend & APIs: FastAPI, Flask, REST APIs, Postman

Database & ETL: PostgreSQL, MongoDB, Airflow, ETL Pipelines, Stored Procedures

DevOps & Tools: CI/CD, Unit & Integration Testing, Agile/Scrum, Matplotlib, Seaborn

EDUCATION

MASTERS IN COMPUTER SCIENCE
Portland State University

Sep 2024 - Present
GPA - 3.75

BACHELOR OF TECHNOLOGY IN INFORMATION TECHNOLOGY
Anna University

Jun 2020 - May 2024
GPA - 3.5

EMPLOYMENT HISTORY

SOFTWARE ENGINEER INTERN
ATTENTION.AD

Jun 2025 - Aug 2025
California, USA

- Developed scalable backend features in **Python** and **SQL** to support real-time data processing and API integrations for ad performance analytics.
- Implemented **CI/CD** pipelines using GitHub Actions and Docker, reducing deployment time by **40%** and increasing release stability.
- Integrated **AI models into data pipelines** for anomaly detection and data quality checks, reducing manual review effort by **60%**.
- Designed **ETL** workflows using Airflow and Python to ingest and transform structured data, improving report generation speed by **35%**.

TECHNOLOGY PARTNER
VIZDALE

Jun 2022 - Feb 2024
Chennai, India

- Designed and maintained backend systems using **PostgreSQL** and **Python**; optimized **SQL** queries and handled schema migrations to boost system performance by **25%**.
- Created Python-based automation pipelines and integrated **ML models** for text classification, reducing manual processing time by **30%**.
- Implemented **Dockerized deployments and FastAPI services**, ensuring scalable and reliable applications.
- Participated in **Agile Scrum** processes by managing 2-week sprints, writing user stories, and improving team delivery speed by **20%** through better coordination.

SOFTWARE DEVELOPER INTERN
RR DONNELLEY INDIA OUTSOURCE PVT LTD

Sep 2023 - Oct 2023
Chennai, India

- Built **Python** automation modules that integrated with a **Scikit-learn-based** classification model to categorize large volumes of financial documents, reducing manual tagging by **20%**.
- Designed relational data schemas and wrote **SQL** stored procedures to feed analytics dashboards, including outputs from **ML models** such as clustering (KMeans) and entity extraction pipelines.
- Developed **REST APIs** to serve ML model outputs (e.g., text classification and OCR-based extraction) as part of a document processing pipeline; validated API functionality using **Postman** and **curl**.

PROJECTS

AI-BASED BUSINESS CARD DIGITIZATION SYSTEM

- Developed an **AI-powered** business card scanner using **Python**, **Optical Character Recognition (OCR)**, and **Natural Language Processing (NLP)** to extract and validate structured contact data from unstructured inputs.
- Engineered a **machine learning** pipeline for entity recognition and data standardization, improving extraction accuracy and reducing manual correction.
- Built an interactive interface to parse, edit, and export contacts in CSV and vCard formats, enabling seamless data integration and enhancing user workflow automation.

INTELLIGENT DOCUMENT PROCESSING (IDP) WITH RAG

- Designed and implemented an **AI-driven** document intelligence system to extract and organize information from semi-structured documents (invoices, contracts, forms) using **Optical Character Recognition (OCR)**, **Natural Language Processing (NLP)**, and **Retrieval-Augmented Generation (RAG)**.
- Achieved over **95%** accuracy in key-value pair extraction through fine-tuned **NLP** models and post-processing pipelines, reducing manual document review by **70%**.
- Deployed scalable microservices using **FastAPI** and **Docker**, integrated **CI/CD** pipelines with **GitHub Actions** to ensure robust, production-ready performance.

MATERIAL RECOMMENDER SYSTEM

- Built a **Python-based** content recommender system using **Scikit-learn** to suggest optimal materials (steels/plastics) based on product requirements like durability, resistance, and temperature.
- Processed a custom dataset of 15k+ entries across 40+ material types using **Pandas** for cleaning and feature engineering, achieving **92.4%** accuracy and reducing material selection time by **30%**.